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Question Bank (QB) – 2024 Pattern
Course: Operating Systems (PCC-203 COM)
Program: SE Computer Engineering
Semester: III
Total Marks: 70
Prepared by: Subject Expert

UNIT I – Introduction to Operating Systems

Sr. No.	CO	BT Level	Question
1	CO1	1	Define Operating System. Explain its objectives and functions.
2	CO1	2	Explain OS as a resource manager and control program.
3	CO1	3	Differentiate between Batch, Multiprogramming, Time-Sharing, and Real-Time OS with examples.
4	CO1	2	Explain the layered structure and microkernel architecture of an OS.
5	CO1	3	Describe different types of system calls. List those related to process control and file management.
6	CO1	3	Explain major components of Linux OS with a neat diagram.
7	CO1	2	Differentiate between Network OS and Distributed OS.
8	CO1	2	Explain kernel and user modes. Why is mode switching necessary.

UNIT II – Process & Thread Management

Sr. No.	CO	BT Level	Question
1	CO2	1	Define process and thread. Explain process state transition diagram.
2	CO2	2	What is a PCB? Describe its role and fields.
3	CO2	3	Compare preemptive and non-preemptive scheduling with examples.
4	CO2	3	Explain various CPU scheduling algorithms (FCFS, SJF, RR, Priority).
5	CO2	4	Consider 4 processes with the following burst times and arrival times. Compute average waiting time and turnaround time using FCFS and SJF (non-preemptive). P1: AT=0, BT=5; P2: AT=1, BT=3; P3: AT=2, BT=8; P4: AT=3, BT=6.
6	CO2	4	Five processes with burst times 3, 5, 2, 5, 5 arrive at t=0. Apply Round Robin scheduling with time quantum=2 ms. Draw Gantt chart and calculate average waiting time.
7	CO2	2	Explain context switching and its impact on system performance.
8	CO2	3	Describe inter-process communication (shared memory and message passing).

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UNIT III – Process Synchronization & Deadlocks

Sr. No.	CO	BT Level	Question
1	CO3	2	Explain critical section problem. Describe requirements for mutual exclusion.
2	CO3	3	Explain binary and counting semaphores with suitable pseudo-code.
3	CO3	3	Describe Peterson's solution for two-process synchronization.
4	CO3	2	Write short notes on Producer–Consumer and Dining Philosopher problems.
5	CO3	2	Define deadlock. State and explain Coffman's conditions.
6	CO3	4	A system has 3 resource types A, B, and C with total instances: A=10, B=5, C=7. The current Allocation and Max matrices for five processes P1..P5 are: P1(0,1,0)/(7,5,3), P2(2,0,0)/(3,2,2), P3(3,0,2)/(9,0,2), P4(2,1,1)/(4,2,2), P5(0,2,1)/(5,3,3). Compute Available and Need matrices. Using Banker's algorithm, determine whether the system is in a safe state and find one safe sequence. If P1 requests (0,0,2), check if it can be granted immediately.
7	CO3	3	Explain various methods of deadlock handling (prevention, avoidance, detection).
8	CO3	2	Distinguish between Mutex, Semaphore, and Monitor.

UNIT IV – Memory Management

Sr. No.	CO	BT Level	Question
1	CO4	2	Define fragmentation. Differentiate between internal and external fragmentation.
2	CO4	3	Explain contiguous and non-contiguous memory allocation methods.
3	CO4	2	Differentiate between paging and segmentation.
4	CO4	3	Explain memory allocation strategies (First Fit, Best Fit, Worst Fit).
5	CO4	2	Define virtual memory. Explain demand paging with page fault handling.
6	CO4	4	Given reference string: 1,2,3,4,1,2,5,1,2,3,4,5 and frame size=3. Compute page faults using FIFO and LRU algorithms.
7	CO4	4	For reference string A,B,C,D,A,B,E,A,B,C,D,E and 3 frames, calculate page faults using Optimal and LRU algorithms.
8	CO4	3	Explain the concept of thrashing and working set model.

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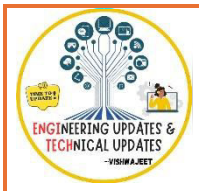
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UNIT V – File System, Storage & I/O Management

Sr. No.	CO	BT Level	Question
1	CO5	2	Explain the structure and components of a file system.
2	CO5	3	Differentiate between file allocation methods: contiguous, linked, indexed.
3	CO5	2	Explain FAT, NTFS, and EXT file systems.
4	CO5	3	Explain various directory structures used in modern OS.
5	CO5	2	Discuss protection and security mechanisms (ACLs, capability lists).
6	CO5	4	Consider disk request queue: 82,170,43,140,24,16,190. Disk head starts at 50. Calculate total head movement for FCFS and SSTF algorithms.
7	CO5	4	Using same data as above, compute total seek time using SCAN and C-SCAN algorithms.
8	CO5	3	Explain programmed I/O, interrupt-driven I/O, and DMA with diagrams.

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Study material provided by: Vishwajeet Londhe

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